Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/26/2007
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Project Title:	Large Scale Fire Dynamics In Spacecraft In Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20899-8665	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-B
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/06) NOTE: Added to Task Book in 10/2006 per GRC info		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McGrattan, Kevin (NIST) Forney, Glenn (NIST)		
Grant/Contract No.:	NNC04IA17I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	With longer-duration human exploration missions to Mars, the moon, or on the International Space Station, the chances for an accidental fire increase. Despite much excellent and useful research to understand the behavior of laboratory flames in microgravity, virtually no work has been conducted to understand large fires in microgravity (?g). This undoubtedly results from the danger and difficulty of conducting such experiments. We propose to use detailed numerical modeling to understand large fires in the reduced gravity conditions (0 gn, 1/6 gn, and 1/3 gn) of a range of extraterrestrial missions (gn is used to represent the acceleration due to normal Earth gravity) . Recently, computational fluid dynamics (CFD) numerical simulation codes based on Large Eddy Simulation (LES) have emerged as excellent tools for understanding large fires in 1 gn. One such program, the National Institute of Standards and Technology (NIST) Fire Dynamics Simulator (FDS), has increased in use world-wide by researchers, fire protection engineers, and others. The range of applications and the sophistication of its underlying engineering models are increasing at a rapid pace. Hence, it seems like an excellent time to start to adapt the code for understanding large fires in reduced gravity, and use this valuable tool to provide the first insights into the fundamental behavior of such fires. This project brings together leading researchers in developing 1 gn fire models with those in microgravity combustion research to examine and modify the sub-grid scale models in FDS for applying it to understanding large fires in reduced gravity. The modified code will then be used to determine the important parameters in spacecraft large-fire growth and spread, suppression, and detection, and for fire-fighting training and fire recreation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Full-scale fire modeling is a very powerful tool for promoting fire safety in space. It facilitates the exploration of space on longer duration missions to the moon or Mars since it allows a better understanding of the microgravity behavior of large fires—which increase in likelihood as the time in space increases. This knowledge will allow improved methods for the prevention of large fires in microgravity and for suppressing such fires if they occur. An ability to model large-scale fires in space vehicles and habitats will allow improved fire safety through: 1.) improved fire detection, 2.) better characterization of materials flammability in reduced gravity, and 3.) improved fire suppression under microgravity conditions. Improvements in the modeling capability will also lead to similar benefits for fire modeling in normal Earth gravity. Human survival in space requires fire-safe design as well as an appropriate response in the event of an unplanned fire. A numerical code capable of describing fire dynamics in reduced gravity will aid greatly in these activities, allowing engineers to design for the actual performance of the materials and system during the fire, rather than relying only on prescriptive codes (which are currently based on 1 g qualification tests for materials to be used on spacecraft). An accurate reduced gravity fire dynamics code can also serve as the basis for a virtual reality training aid to acquaint astronauts to the behavior of large fires in space. These techniques are in development for 1 g application; however, they are of particular value for space applications since full-scale training with real fires is not practical. Finally, there are many materials that fail the National Aeronautics and Space Administration (NASA) flammability screening tests but are allowed onboard space craft via various waivers (which limit the times and ways that they can be present in the crew compartment). It would be very useful to model the behavior of these items if accidentally exposed to an ignition source.
	Progress was made in five distinct areas. Specifically, we examined: 1. Oxygen limitations to fire growth in reduced gravity environments; 2. FDS predictive capability for solid material burning rates in normal and reduced gravity; 3. Overall fire behavior in space habitation modules, and 4. Extinction in co-flow diffusion flames. Finally, we also undertook measurements and analysis to assess the use of 5. Flame size as a method to determine the heat release rate from a flame. 1. Oxygen Limitations to Fire Growth in Reduced Gravity Environments This work investigated two aspects of the problem of predicting fires in variable gravity: 1.) validation of the sub-grid models of turbulence in FDS for conditions similar to those expected to be encountered in spacecraft; and 2.) exploration of the global effects of gravity and ventilation on the potential heat release from a fire. The goal of the first part of the work was to validate the capability of FDS to predict the movement of the turbulent air streams in micro- and variable-gravity environments. Unfortunately, experimental data on ventilation flows in variable gravity are not yet available. Nonetheless, there are experimental data in the heating and ventilation literature for isothermal, low speed air flow in simple geometries; these are relatively close to the conditions expected for low-gravity vehicles and habitation modules. Using an available dataset on ventilation in a rectangular room (3m x 3 m x 9 m) with variable inlet and exit locations and air flows, FDS was used to predict the velocities for comparison with the data. For a range of vertical and axial locations in the room, with air flows of -0.2 m/s to 0.4 m/s, FDS predictions of the average gas velocities were in reasonable agreement with the measurements. The significance of the agreement goes beyond gas velocities since the turbulent motion which affects the development of the velocity profile is the same physical phenomena which affects mixing, which is crucial for fire growth and spread. The goal of the second part of this work was to understand how the supply of fresh, unburned air to the fire location is affected by variable gravity and ventilation flow in a module. The calculations examined one question: Under what conditions will a large fire in low gravity be non-viable due to limitations on the transport of oxygen to the flame? The calculations indicate that for uniform, co-flowing, laminar inlet flow, and fuel sources of representative size (r=5 cm to r=15 cm), the heat release rate can be limited by the supply of air. For higher air flow and smaller flames, the flames reach near peak values after about 20 s, whereas for larger flames with lower airflow, the heat release can be oxygen limited. Nonetheless, with any level of gravity above 1/6 g , there is sufficient buoyancy driven flow to make the heat release rate comparable to 1 gn results. A most important conclusion, however, is that putting the same burner into a 0 gn module with representative turbulent flow provides sufficient air so that the flames behave much more like flames in 1 gn than laminar co-flow flames in 0 gn. 2. FDS Predictive Capability for Solid Material Burning Rates in Normal and Reduced Gravity The above work on oxygen limitations to fire growth in low gravity used a fire of prescribed fuel addition rate. A real fire, however, typically depends upon the feedback of the heat from the fire to decompose the solid fuel into gaseous compounds which then burn in the flame. Hence, to predict fire growth and spread (in 1 gn or low-gn), FDS must be able to properly account for the decomposition and burning of solid-phase materials. While it is desired to make such comparisons between model prediction and experiment based on samples of representative scale, no such data for

Task Progress:	materials burning in 0 gn or low-gravity are available. Hence, as a first step, 1 g burning rate data were generated for intermediate-scale samples using the NIST cone calorimeter (a device for irradiating a 10 cm x 10 cm 2.5 cm thick polymer sample and measuring the burning rate), and FDS was used to predict the burning rate. This approach provides some measure of the model's capacity for predicting fire growth, since an ability to predict the burning of such a simple configuration is a clear prerequisite for modeling large fires (which ultimately will also involve burning on the scale of cone samples). The goal of this work then was to assess the relative importance of various numerical, physical, and experimental parameters on the predicted burning of the sample. It was found that to accurately predict the heat release rate of the burning polymer, it was important to properly specify the domain size and select a fine enough grid resolution. The material properties: heat of vaporization, ignition temperature, and the activation energy for decomposition all needed to be carefully specified, and some experimental configuration factors (e.g., lip on sample holder and presence of the cone heater) also needed to be accurately described. To get correct values for cases with low imposed radiant heat flux, the flame heat transfer to the sample needed careful attention. Of minor importance were the use of direct numerical simulation rather than large-eddy simulation, and use of 3-D instead of 2-D coordinates in the calculation method. Presence of either exhaust flow or backside insulation had little affect on the calculations. A major finding was that the empirically determined effective heat of gasification was the most important parameter necessary for accurate prediction of the burning rate, and that values of the heat of gasification extracted from the cone measurements did not yield a predicted heat release rate in agreement with those measured in the cone. 3. Overall Fire Behavior in Space Habitation Modules. Work in this area focused on predicting the Mir fire and understanding any fire risk from it, and starting to understand how fires in an ISS module (as a representative case) might differ from a normal gravity room fire. First we used ground-based tests of a similar oxygen generator to estimate that the heat output for the Mir fire was about 5 kW, in a volume of about 0.2 m3. We tried to get the numerical code to simulate a 5 kW fire burning in 0 gn, but various limitations in version 4.0 of FDS did not allow us to achieve that (and still have the radiation transport part of the calculation working). Hence, we must wait for FDS version 5.0 to be released to continue the investigations. In lieu of simulating a fire itself, we performed equilibrium calculations to estimate the product gas temperatures downstream of the oxygen generator to start to get a sense of the fire threat from the burning stainless steel of the oxygen generator. The final product temperature, based on stoichiometric burning of iron with O2, is about 3040 K. Nonetheless, if the mixing fan flow (1333 L/min) is combined with these high-temperature products, the final temperature is near 500 K We then attempted to simulate the radiation from the Fe2O3 particles, but again, this was not possible with FDS 4.0. Our conclusion at this point was that if the mixing fan were operating, the fire threat from the hot product gases would be low; however, if hot glowing particles were present, they could be an ignition source. 4. Extinction in co-flow diffusion flames. Eventually, FDS will need to have a modified extinction model for flames in reduced gravity. In support of that goal, we collaborated with F. Takahashi at NASA-GRC in work to understand stabilization and extinguishment of co-flow diffusion flames (formed over cup-burners) in normal and reduced gravity. Progress was made in numerical modeling of the influence of chemical inhibitors, flame oscillations, and fuel type on flame extinguishment, and how these change with gravity levels. For example, the limiting temperature for shutdown of the heat releasing reactions was found to be around 1300 K in 0 gn, vs. 1700 K in 1 gn. The next step is to try to generalize the findings and then incorporate the effects into large fire models. 5. Flame size as a method to determine the heat release rate from a flame. A few experiments slated to fly on the International Space Station concern solid sample combustion. The hardware has already been built, but the researchers now feel that getting the value of the heat release of the burning sample would be of use. In order to allow the researchers to extract heat release rate data from their experiments, we developed an experimental technique for simultaneously obtaining the flame size, heat release and smoke point for flames over condensed-phase materials. From these, a correlation can be determined so that the heat release rate can be obtained solely from the optically measured flame area. The data show that recorded images of the luminous flame (from soot emission) and that from CH emission (approximating the stoichiometric contour) diverge above the smoke point fuel addition rate so that the smoke point is readily determined. The flame area from the stoichiometric contour are superior to the flame height based on the luminous flame for correlating heat release rate. In addition, there exist several important considerations with regard to the actual implementation of the technique in a screening apparatus. For the small flames used, account must be made of the unsteady and the multi-dimensional heat loss terms in the energy balance equation, since these can greatly reduce the effective heat input to the sample from the radiant source.
Bibliography Type:	Description: (Last Updated: 11/01/2007)
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